

(Some) Vector & tensor algebra operations

1. Transpose

If $\underline{\underline{T}} = T_{ij}$ then $T_{ji} = \underline{\underline{T}}^t$ [understood as $(\underline{\underline{T}}^t)_{ij}$ entry]

Remember: flipping the indices creates a *new* tensor (“the transpose”); $T_{ij} = T_{ji}$ *only* for *symmetric* tensors!
(**Note:** Panton denotes transpose with t , but you might also see $^\top$.)

2. Multiplication

2.1. vector/vector (dot product; row vector $\times$ column vector = scalar) – *only one that commutes*

$$\underline{u} \cdot \underline{v} = u_i v_i = v_i u_i = \underline{v} \cdot \underline{u}$$

Note: For $\underline{v} \cdot \underline{v} = v_i v_i$ **don’t** write v_i^2 because this expression has a free index, and it’s therefore a vector!

2.2. vector/tensor (row vector $\times$ matrix = row vector, or matrix $\times$ column vector = column vector) – *does not commute*

(**Note:** Panton puts a $\cdot$ between vectors and tensors for clarity; optional when using 1 vs. 2 underlines.)

$$\underline{v} \underline{\underline{T}} = v_i T_{ij} \quad \text{or} \quad (\underline{v} \underline{\underline{T}})_{j\text{th entry}} = \sum_{i=1}^3 T_{ij} v_i$$

but we’re allowed to (carefully!) re-index as

$$(\underline{v} \underline{\underline{T}})_{i\text{th entry}} = \sum_{j=1}^3 T_{ji} v_j$$

which **does not commute** because

$$\underline{\underline{T}} \underline{v} = v_j T_{ij} \quad \text{or} \quad (\underline{\underline{T}} \underline{v})_{i\text{th entry}} = \sum_{j=1}^3 T_{ij} v_j$$

Note: If we always write T_{ij} , then you can remember what index to put on v by whether $\underline{v}$ is in-front of $\underline{\underline{T}}$ (then its v_i , **and $\underline{v}$ is a row vector**) or behind $\underline{\underline{T}}$ (then its v_j , **and $\underline{v}$ is a column vector**) in the symbolic notation expression.

The following useful fact follows: $\underline{v} \underline{\underline{T}}$ is the row vector with the same entries as the column vector $\underline{\underline{T}}^t \underline{v}$.

2.3. vector/vector (dyadic product; column vector $\times$ row vector = matrix) – *does not commute*

$$\underline{u} \underline{v} = u_i v_j$$

does not commute because

$$\underline{v} \underline{u} = v_i u_j = u_j v_i$$

But, if we let $\underline{\underline{T}} = \underline{u} \underline{v}$, then we have shown that $\underline{v} \underline{u} = \underline{\underline{T}}^t$.

Sometimes we denote “ $\underline{u} \underline{v}$ ” as “ $\underline{u} \otimes \underline{v}$ ” (so we don’t accidentally confuse it for “ $\underline{u} \cdot \underline{v}$ ”).

2.4. tensor/tensor (direct product; matrix $\times$ matrix = matrix) – *does not commute*

$$\underline{\underline{R}} = \underline{\underline{T}} \underline{\underline{S}} \quad \text{or} \quad R_{ij} = T_{ik} S_{kj} \quad \Rightarrow \quad (\text{for example}) \quad T_{ik} S_{jk} = \underline{\underline{T}} \underline{\underline{S}}^t \neq \underline{\underline{R}}!$$

This is simply multiplication of matrices, where k runs over the i th row of $\underline{\underline{T}}$ and j th column of $\underline{\underline{S}}$:

$$R_{ij} = \sum_{k=1}^3 T_{ik} S_{kj}$$

“Row times column” $\Rightarrow$ inner indices (2nd one on T and 1st one on S , in index notation) must match!

2.5. tensor/tensor (double contraction; “dot product” for tensors) – *does not commute*

$$\underline{\underline{T}} : \underline{\underline{S}} = T_{ij} S_{ji} \quad \Rightarrow \quad (\text{for example}) \quad T_{ij} S_{ij} = \underline{\underline{T}} : \underline{\underline{S}}^t \neq \underline{\underline{T}} : \underline{\underline{S}}$$

Note: If we define $R_{kl} = T_{kj} S_{jl}$ then $\text{tr}(\underline{\underline{R}}) = R_{kk} = R_{ii} = T_{ij} S_{ji}$, in other words $\underline{\underline{T}} : \underline{\underline{S}} = \text{tr}(\underline{\underline{T}} \underline{\underline{S}})$.